

Lab 3 — Architectural Justification

Digital Art Commission & Watermarking Portal | Problem Statement #58

PES University — Dept. of CSE • SRN: PES1UG24CS593 • Name: _____ Date: _____

Architecture Selection: We chose the **Layered Architecture** for the Digital Art Commission & Watermarking Portal System.

1. Architectural style analysis

Style	Strength for this scenario	Why it falls short / costs	Verdict
Microservices	Watermarking is CPU-bursty and would scale well as its own deployable.	FR-005 requires that payment, entitlement and download authorisation agree at all times. Splitting them introduces eventual consistency exactly where an unpaid original must never leak.	Rejected
Client-Server	Simple to deploy for a two-actor portal with centralised data.	Business rules drift towards the client. The fail-closed watermark rule in FR-001 must never be enforced anywhere a caller can bypass.	Rejected
Layered	One transactional data layer, one code path from upload to publication, clean separation of presentation, business rules and asset access.	Layers add call overhead and the whole application scales as one unit; heavy watermarking has to be pushed to an asynchronous worker rather than scaled independently.	Selected

2. Reason one — the payment-to-download rule must be one transaction, not a negotiation

FR-005 states that final source files unlock only after milestone payment is verified, and its acceptance criteria are absolute: before payment every request returns 403 with no file data, and after payment exactly one entitlement exists. That is a consistency requirement, not a throughput one. In the layered design, the Payment Service Component, the Entitlement & Audit Component and the Commission Database Component sit in the same process and the same transaction boundary, so recording a verified payment and creating the entitlement either both happen or neither does. Splitting them across services would put a network boundary in the middle of the one invariant the business cannot afford to get wrong, and would make duplicate entitlements on a replayed gateway callback a real possibility rather than a transactional impossibility.

3. Reason two — one code path may publish a preview, and it is watermarked

FR-001 requires that the Client Buyer never sees an unwatermarked draft, and the flow specification for UC-03 requires the system to fail closed when watermarking fails. A layered structure makes that structural: the Web Portal UI Component cannot reach asset storage at all, because only the Data Access Layer holds the credentials, and every upload must pass through the Watermarking Service Component before the Order Manager will publish a preview. There is one path from upload to publication and it contains the verification step. In a distributed design, any service holding storage credentials could publish directly, and the guarantee would rest on convention rather than on structure.

4. Security advantage

The Asset Store Adapter Component is the only component in the system holding credentials for the private object store, and it lives in the Data Access Layer with no inbound path from presentation. A download therefore cannot be obtained by guessing a URL or by compromising the UI layer: the request must pass the Entitlement & Audit Component, which checks that a verified payment exists for that milestone, before a 60-minute signed URL is minted. Every one of those checks — along with brief, draft, approval and payment actions — is written to an append-only audit table, satisfying the 365-day immutable audit requirement in NFR-002 and giving a defensible record if authorship or payment is later disputed.

5. Performance benefit

NFR-001 requires 95% of authorised download URLs to be generated within 2 seconds at 300 concurrent users. Signing a URL is a local cryptographic operation against a key the Asset Store Adapter already holds, so in the layered design the request path is an in-process call chain — entitlement check, then signature — with a single database read and no network hop between components. The expensive work, watermarking, is deliberately kept off that path: the Watermarking Service Component runs as an asynchronous worker, so an artist's upload returns as soon as the original is stored and image processing never blocks a buyer waiting on a download.

6. Components and interfaces modelled

Component	Source	Responsibility
Order Manager Component	Given	Orchestrates brief, milestone, draft and approval flow across the business layer
Payment Service Component	Given	Calls the external gateway and reports verified payment to the entitlement component
Watermarking Service Component	Identified	Applies the diagonal overlay and verifies it before any preview is published (FR-001)
Entitlement & Audit Component	Identified	Records one entitlement per verified payment and writes immutable audit events (FR-005, NFR-002)
Asset Store Adapter Component	Identified	Holds the only credentials for private storage; mints 60-minute signed URLs (NFR-001)
Commission Database Component	Identified	Single transactional store for commissions, milestones, entitlements and the audit log
Web Portal UI Component	Identified	Presentation layer for both the Client Buyer and the Digital Artist

Interfaces shown in the diagram: *IBuyerPortal*, *IArtistStudio*, *ICommissionAPI*, *IPayment* (Order Manager ↔ Payment Service, payment processing requests), *IWatermark*, *IEntitlement*, *IPersistence*, *IAssetStore* and *ISignedUrl*, plus «use» dependencies to the Payment Gateway, the object store and the append-only audit table.